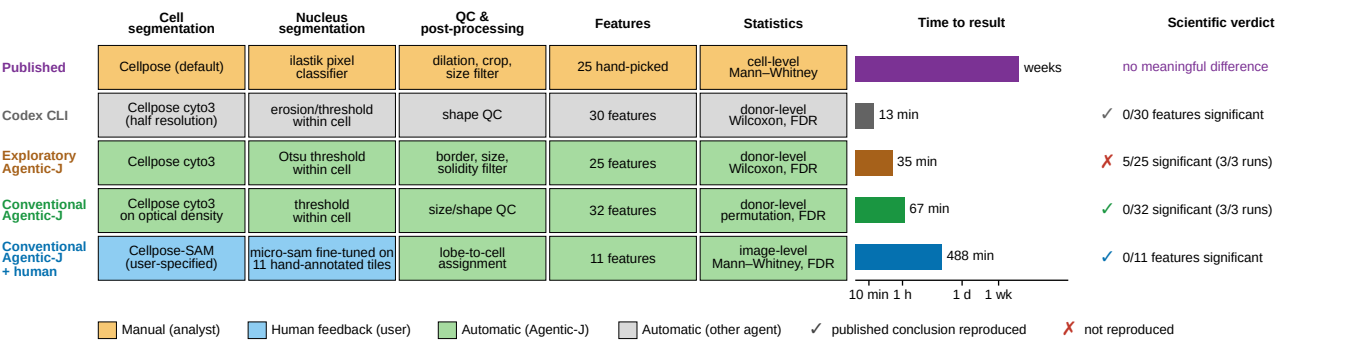
Cell segmentation
Nucleus segmentation
QC & post-processing
Features
Statistics
Time to result
Scientific verdict
Published
Cellpose (default)
ilastik pixel classifier
dilation, crop, size filter
25 hand-picked
cell-level Mann–Whitney
weeks
no meaningful difference
Codex CLI
Cellpose cyto3 (half resolution)
erosion/threshold within cell
shape QC
30 features
donor-level Wilcoxon, FDR
13 min
✓ 0/30 features significant
Exploratory Agentic-J
Cellpose cyto3
Otsu threshold within cell
border, size, solidity filter
25 features
donor-level Wilcoxon, FDR
35 min
✗ 5/25 significant (3/3 runs)
Conventional Agentic-J
Cellpose cyto3 on optical density
threshold within cell
size/shape QC
32 features
donor-level permutation, FDR
67 min
✓ 0/32 significant (3/3 runs)
Conventional Agentic-J + human
Cellpose-SAM (user-specified)
micro-sam fine-tuned on 11 hand-annotated tiles
lobe-to-cell assignment
11 features
image-level Mann–Whitney, FDR
488 min
✓ 0/11 features significant
10 min
1 h
1 d
1 wk
Manual (analyst)
Human feedback (user)
Automatic (Agentic-J)
Automatic (other agent)
✓ published conclusion reproduced
✗ not reproduced